

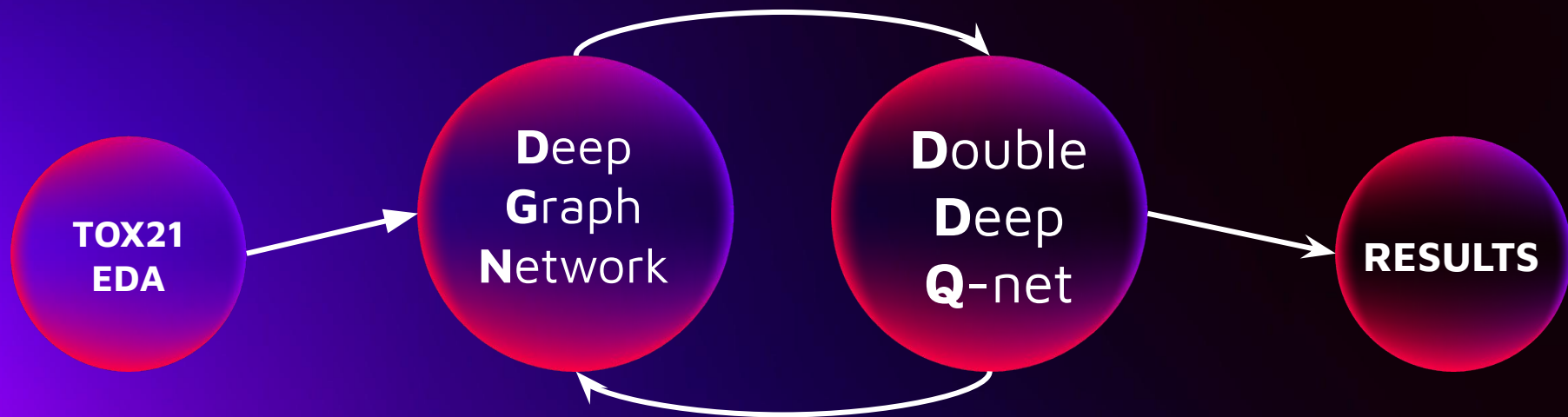
DGN & DDQN Agent

toxicity counterfactual explainability

Lorenzo Benucci - 2219690
Emilio Leo - 1966813



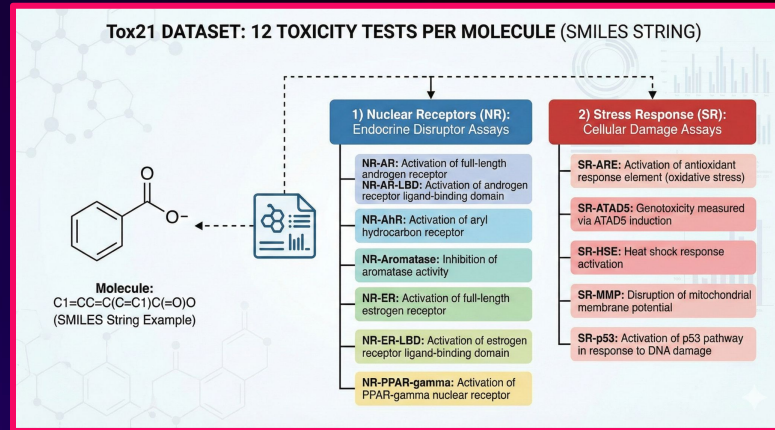
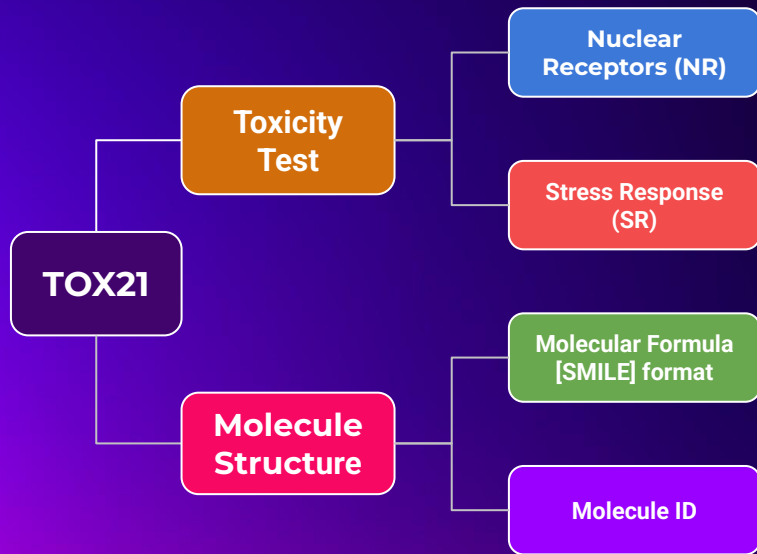
Our Pipeline



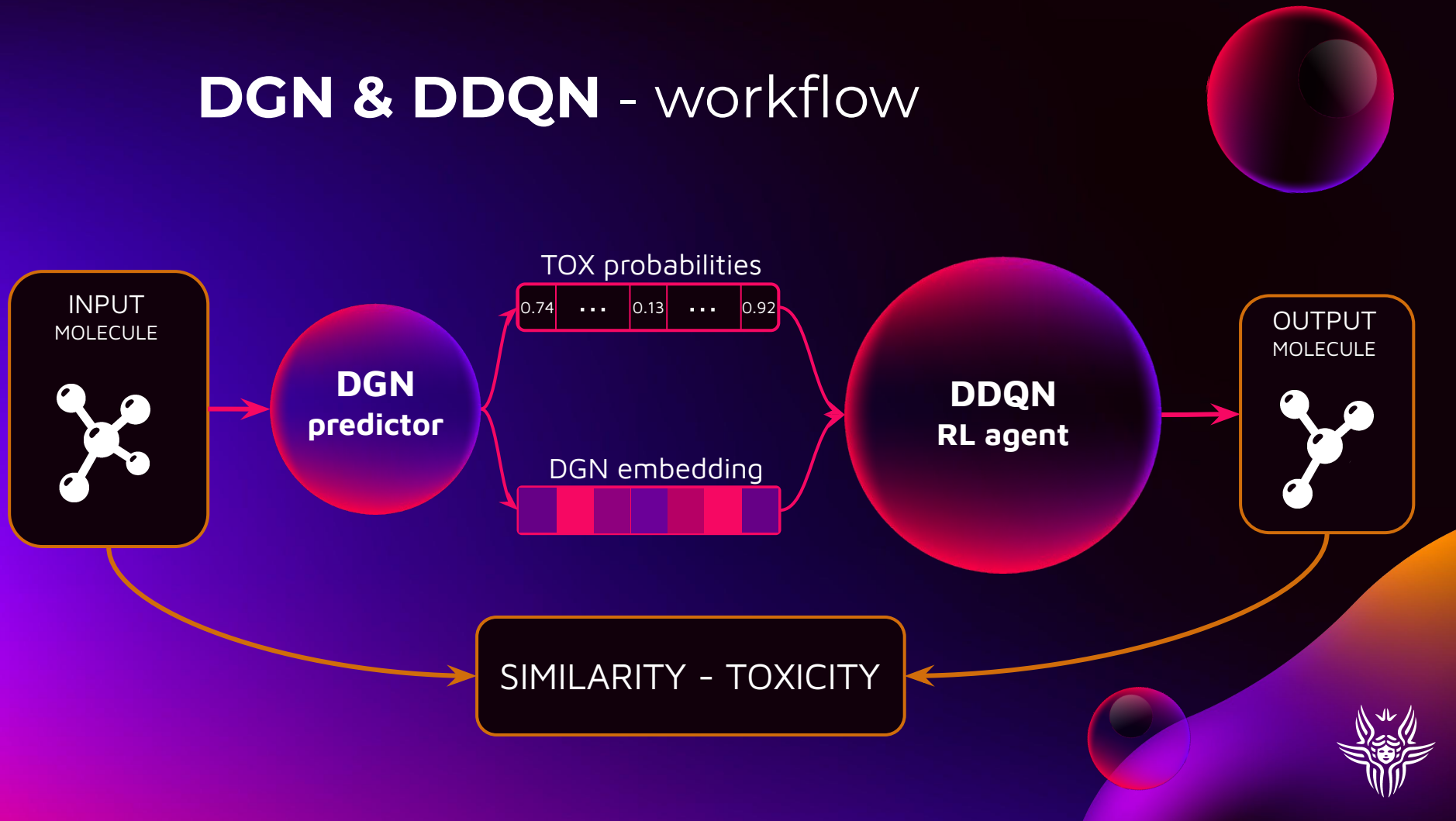
GOAL: flipping toxicity, maximizing similarity



Dataset - TOX21



DGN & DDQN - workflow



DGN - Predictor

- **The Oracle**: predicting toxicity $\rightarrow \left\{ \begin{array}{l} \text{graph} \rightarrow \mathcal{P}_i(x) \\ \varphi_\theta : \mathcal{G} \rightarrow [0, 1]^{12} \end{array} \right.$

Why **DGN**?

- Native non-euclidean form
- $\varphi_\theta(PX, PAP^T) = \varphi_\theta(X, A)$
- Learns high-D representations
- Outputs continuous distribution

our choice: **GINEConv**

$$h_v^{(k)} = \text{MLP} \left((1 + \epsilon) \cdot h_v^{(k-1)} + \sum_u h_u^{(k-1)} \right)$$

MLP

+

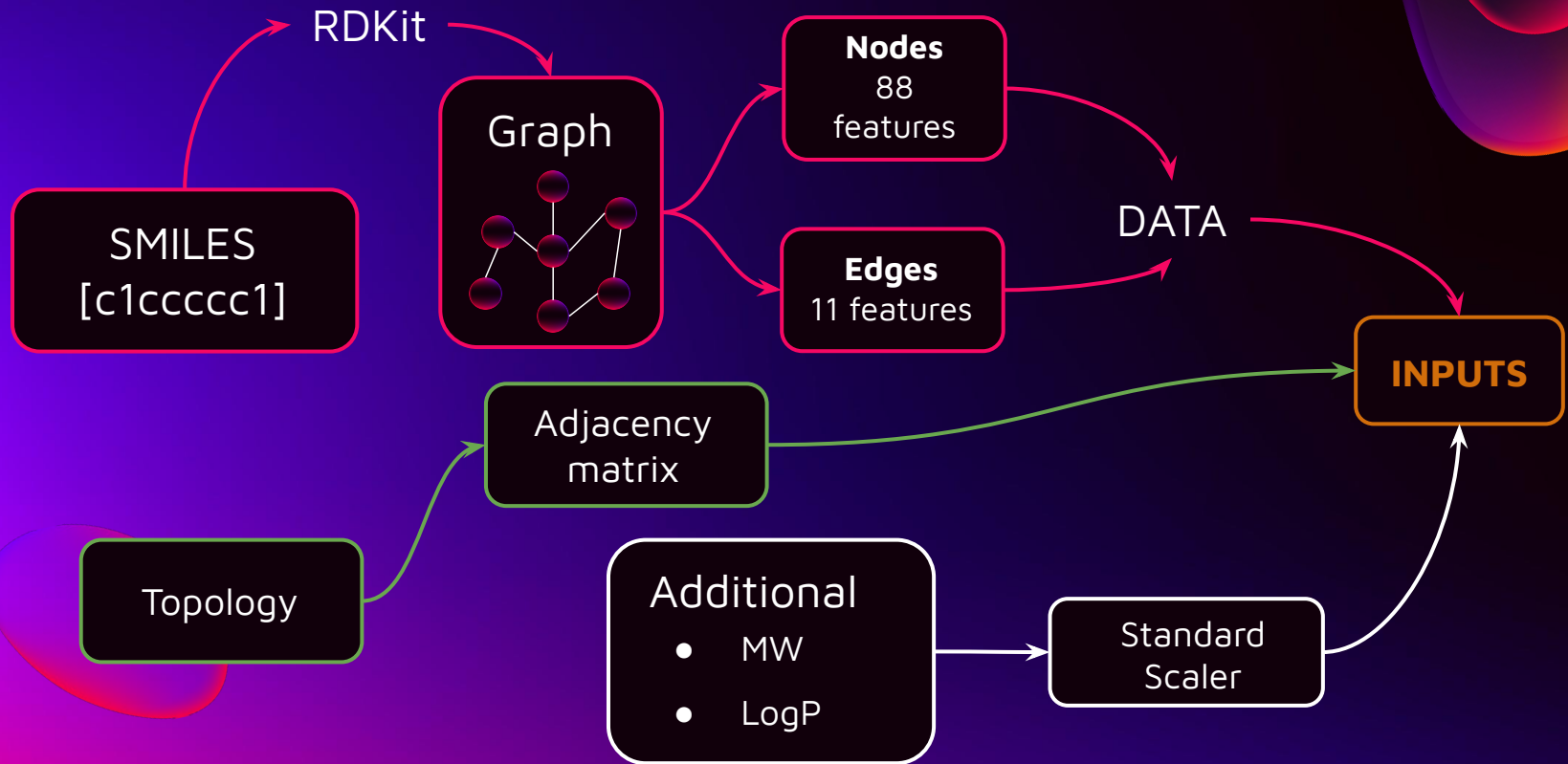
Weisfeiler
Lehman

including
edges

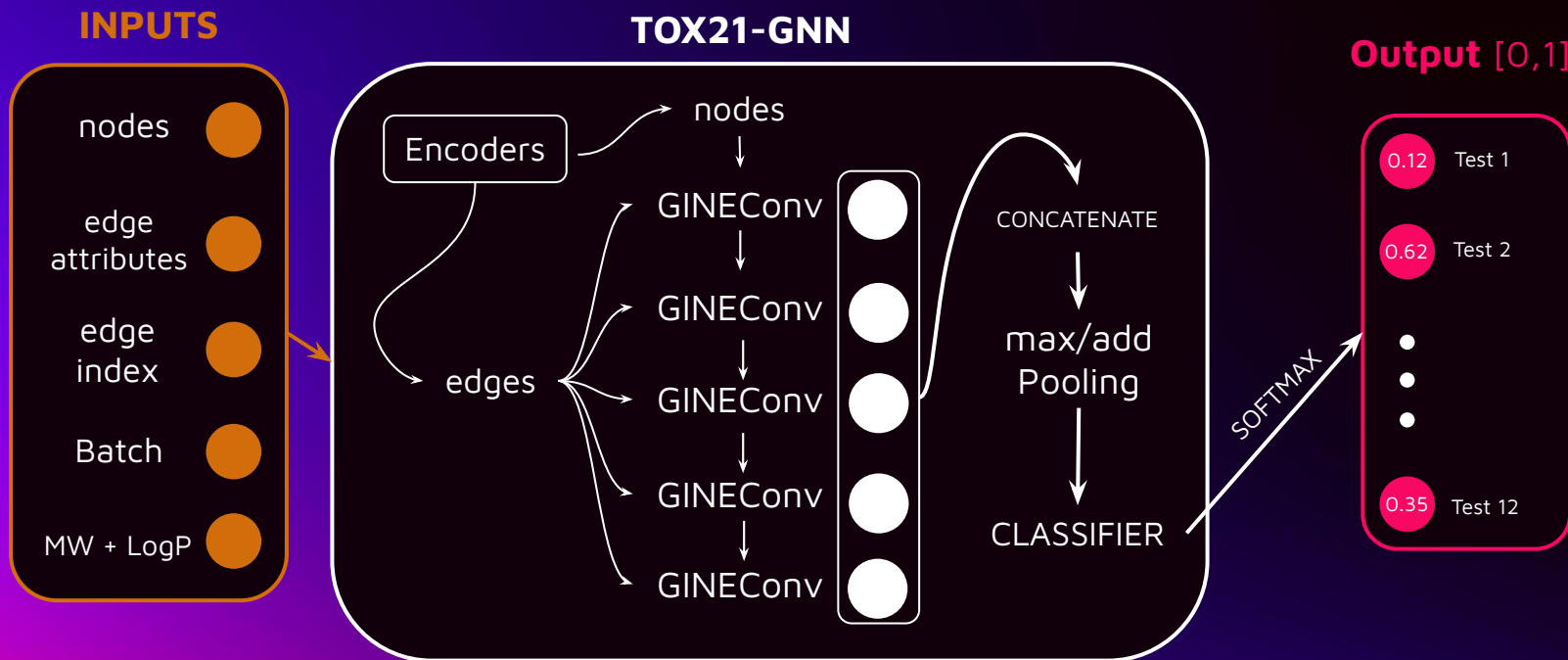
“Agent
Friendly”



DGN - Reading TOX21



DGN - Architecture



DGN - Training

TOOLS

Dynamic Class
Weighting

$$\omega_i = \frac{N_{neg}}{N_{pos}}$$

Masked
backpropagation

$$\mathcal{L} \leftarrow \mathcal{L} \cdot \mathcal{M}$$

Hybrid LR
Scheduler

- Linear Warmup
- Cosine Decay

ROC - AUC

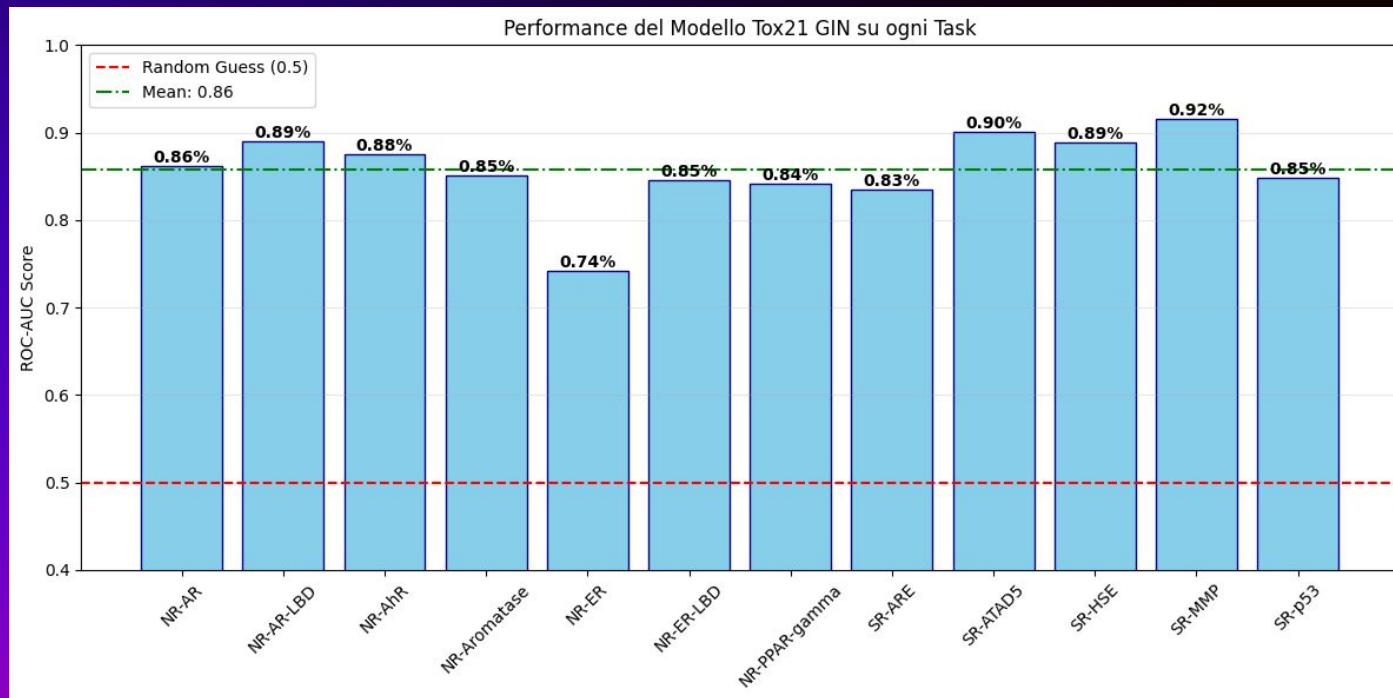
- here more reliable than accuracy
- class-independent computed
- continuous Spectrum reliability

PARAMETERS

- Adam Optimizer
- LR $\approx 6.1e-3$
- Epochs = 200
- Batch Size = 64
- Dropout ≈ 0.41
- W. Decay $\approx 2.6e-4$
- Hidden Ch. = 128
- [T|I|V] = [0.7|0.15|0.15]



DGN - Results*



* performances on test set (15% , N. molecules = 1174)



MEG - Environment

STATE SPACE



GiNEConv output
concatenation

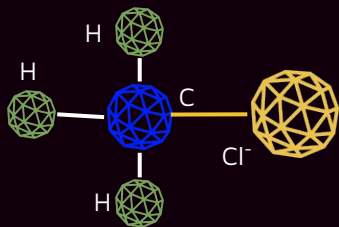


Global Pooling

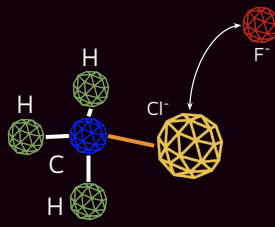


Global Properties

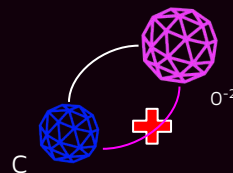
ACTION SPACE



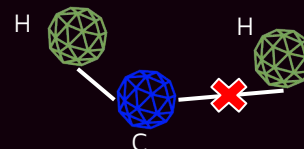
ADD-atom



SUB-atom



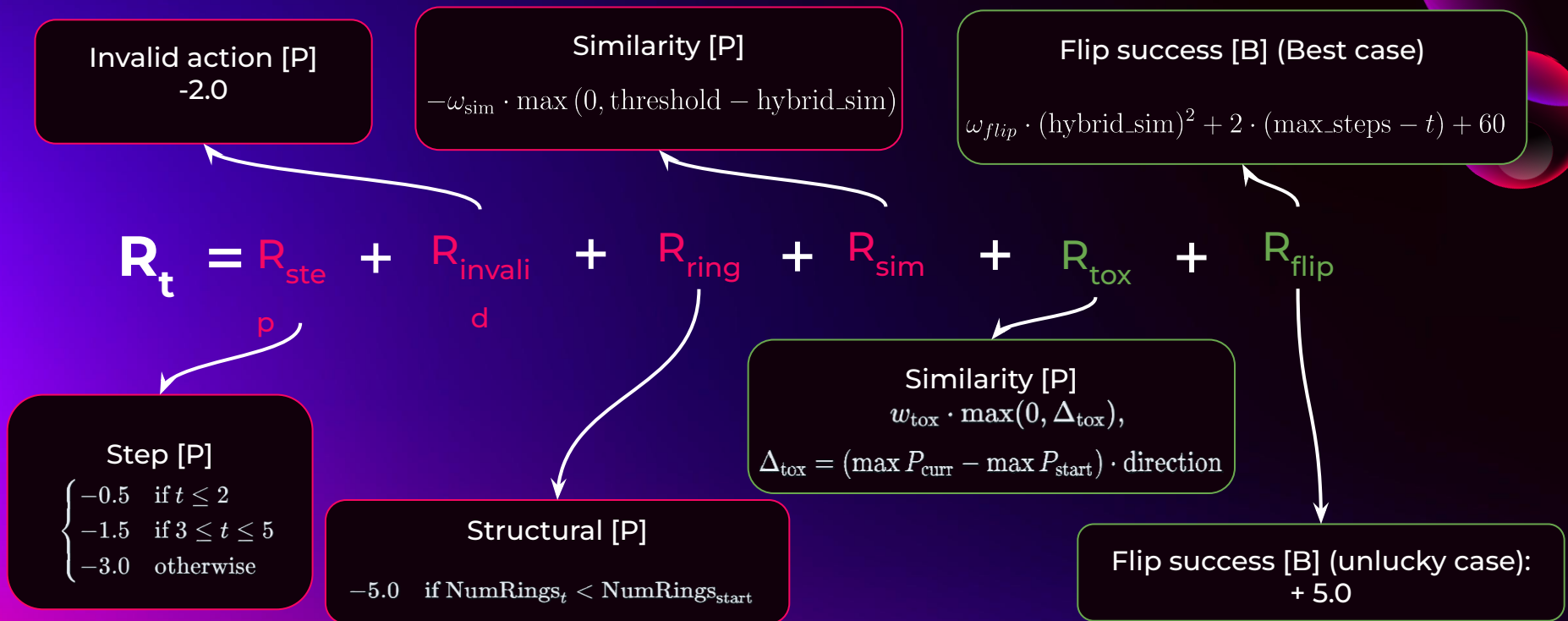
ADD-bond



RM-Bond



DDQN - Reward



DDQN - Agent

DDQN

FEAT LAYERS

Linear(in, h)

LayerNorm(h)

SiLU

Linear($h, h/2$)

LayerNorm($h/2$)

SiLU

VALUE STREAM

Linear($h/2, 1$)

ADVANTAGE STREAM

Linear($h/2, n_{actions}$)

OUTPUT

Estimated Q-value

$$Q(s, a) = V(s) + \left(A(s, a) - \frac{1}{|\mathcal{A}|} \sum_{a'} A(s, a') \right)$$

INPUTS

GNN
embedding



DDQN - Weighted replay buffer

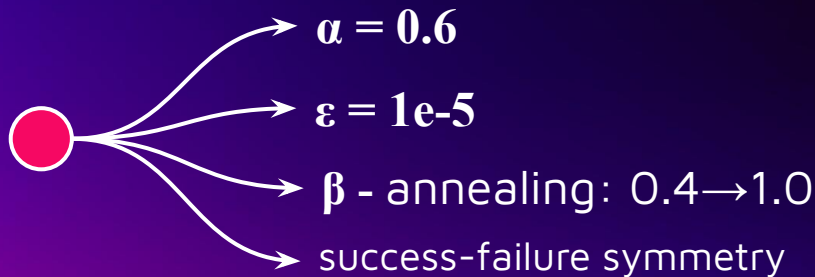


PRIORITY
 $p_i = |R_i| + \epsilon$

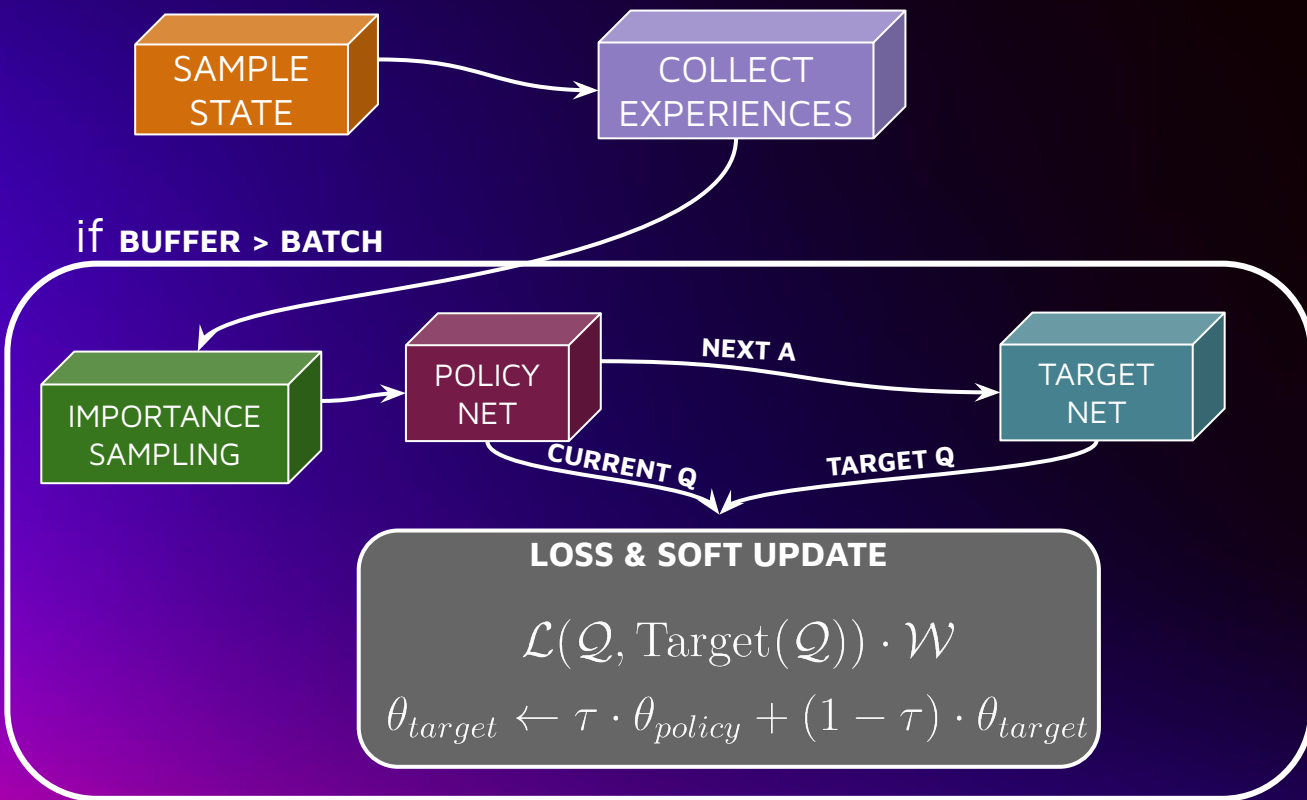
SAMPLING PROB.
 $P(i) = \frac{p_i^\alpha}{\sum_k p_k^\alpha}$

IMPORTANCE SAMPLING
 $w_i = \left(\frac{1}{N \cdot P(i)} \right)^\beta \cdot \frac{1}{\max(\omega)}$

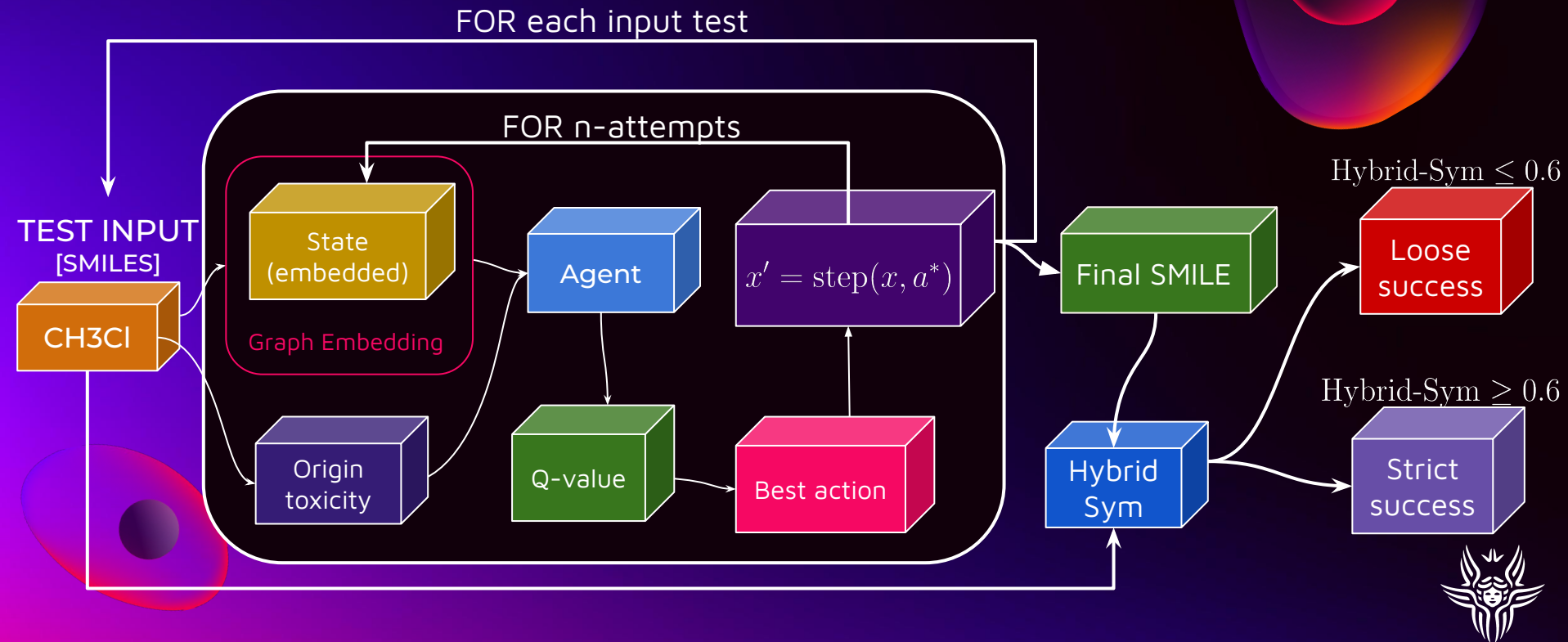
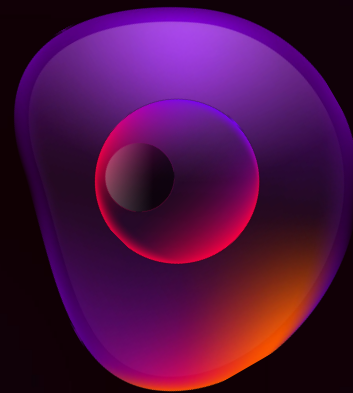
Correction: $\mathcal{L} \leftarrow \mathcal{L} \cdot W$



DDQN - Optimization



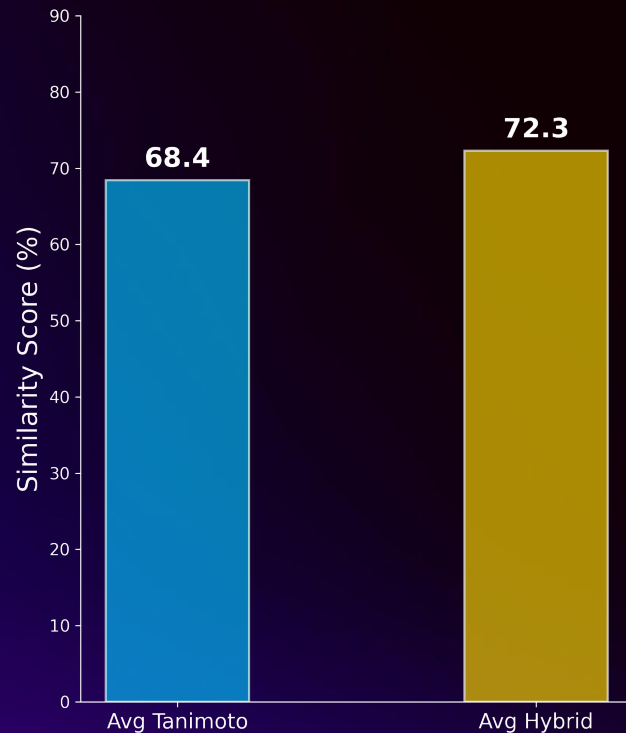
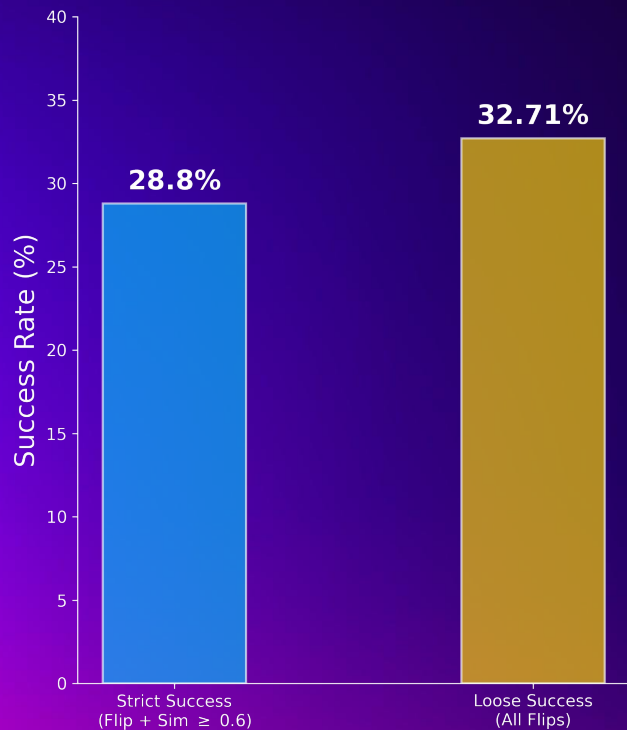
DDQN - Evaluation



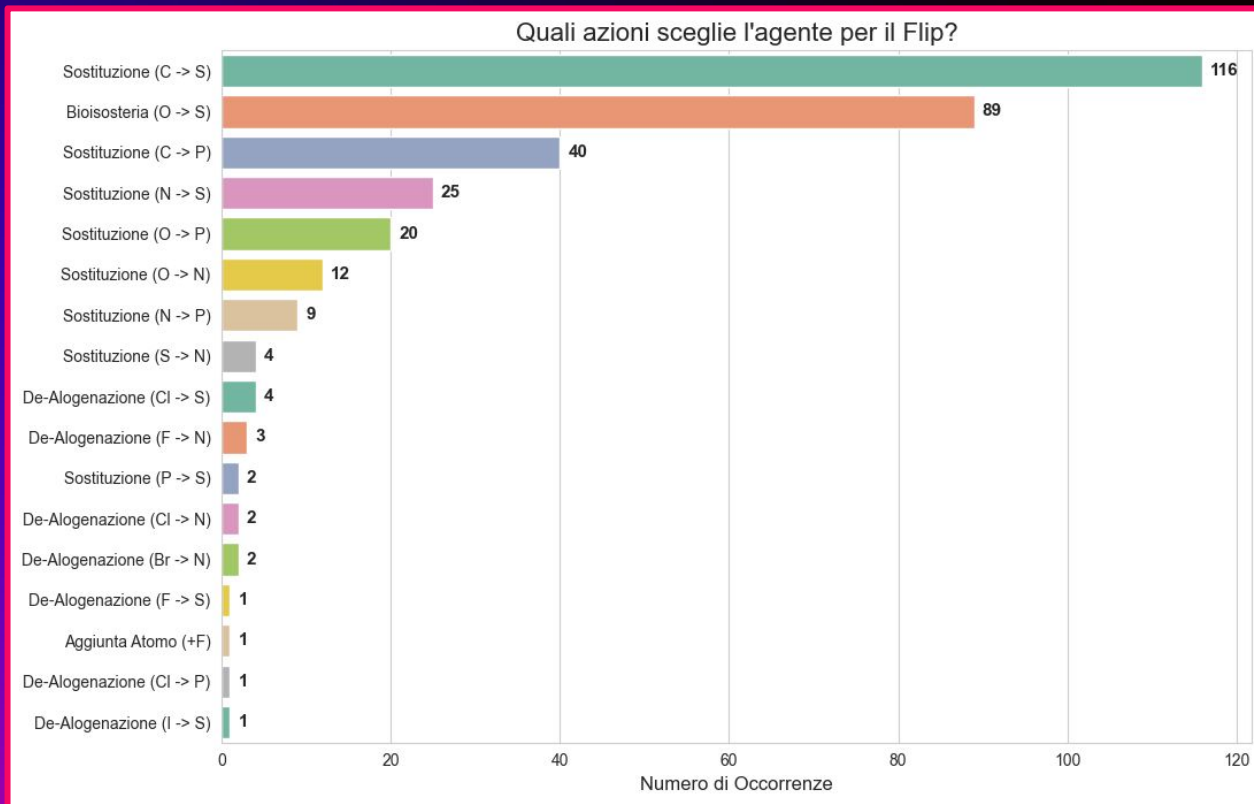
Results - success



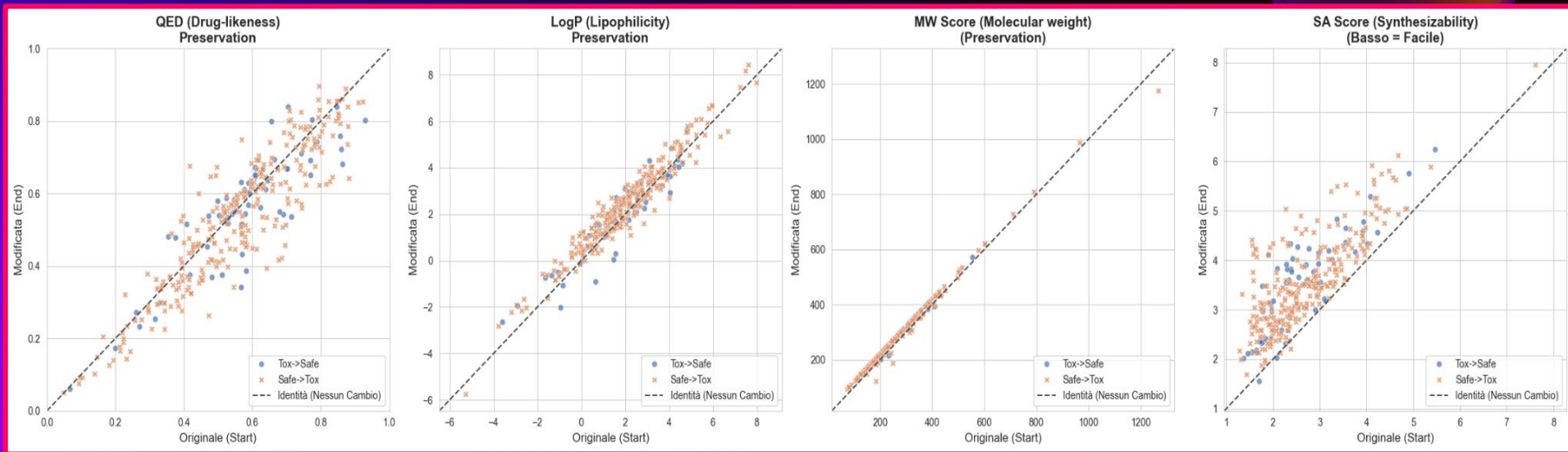
Average Steps = 1



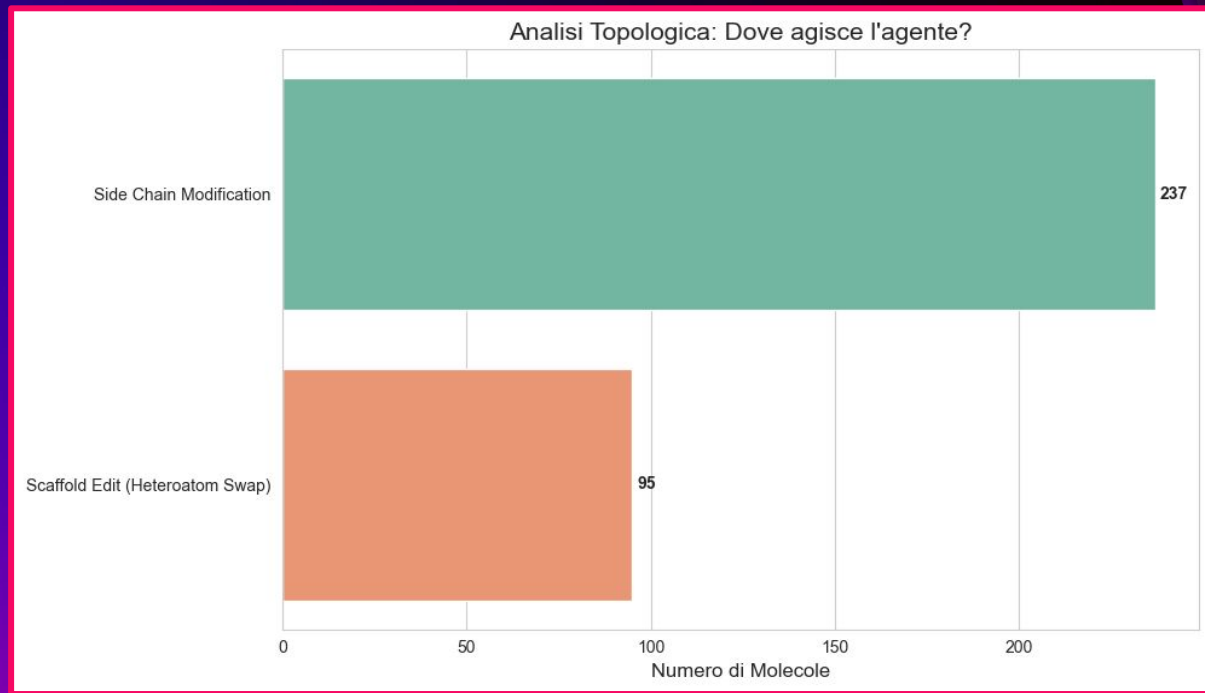
Results - best actions



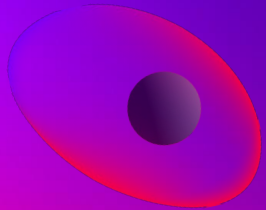
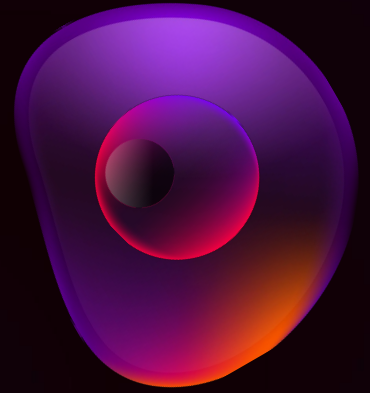
Results - chemical properties



Results - topology



Thank you



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